

BALTIMORE DOWNTOWN, MD, USA

WMO: 745944

Lat: 39.281N

Lon: 76.611W

Elev: 20

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93784

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.8	(c) 20.7	(d) -4.3	(e) 4.4	(f) 19.4	(g) 0.1	(h) 5.5	(i) 23.0	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	95.5	75.6	92.9	74.3	90.5	73.4	78.2	90.3	77.0	88.0	75.9	86.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	130.9	83.2	73.7	125.6	82.3	72.6	121.0	81.8	41.5	90.4	40.2	87.8	39.1	86.3	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	12.8	99.5	4.8	3.7	9.4	102.2	6.6	104.4	4.0	106.4	0.5	109.1
			9.9	80.4	4.3	1.5	6.8	81.5	4.2	82.4	1.8	83.2	-1.3	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	36.4	38.7	46.8	58.1	67.3	76.8	81.2	79.5	72.8	61.1	50.4	41.3	(6)
	DBStd	17.49	9.91	9.04	9.20	8.63	7.92	6.14	5.12	4.75	6.27	8.08	7.96	8.34	(7)
	HDD50	1345	433	330	173	22	0	0	0	0	0	7	90	290	(8)
	HDD65	3858	886	735	566	243	67	3	0	0	10	174	440	734	(9)
	CDD50	4747	12	15	75	264	537	805	968	913	683	351	102	22	(10)
	CDD65	1785	0	0	3	35	140	358	503	448	243	53	2	0	(11)
	CDH74	16909	0	3	28	312	1120	3442	5594	4379	1725	299	6	1	(12)
	CDH80	6643	0	0	4	97	376	1359	2532	1691	515	69	0	0	(13)
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15) Precipitation	PrecAvg	42.0	3.1	3.1	3.9	3.1	3.9	3.1	3.8	4.1	3.6	3.0	3.5	3.7	(15)
	PrecMax	61.6	8.7	6.9	9.8	7.6	9.3	10.0	11.7	14.8	8.8	8.7	8.7	8.1	(16)
	PrecMin	29.6	0.4	0.7	0.9	0.4	0.6	0.6	0.7	0.3	0.1	0.1	0.4	0.7	(17)
	PrecStd	8.8	1.7	1.7	2.1	1.7	2.0	1.8	2.1	3.0	2.4	1.9	1.9	2.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.2	69.5	78.0	88.8	92.9	97.6	100.3	97.0	92.9	86.2	74.3	68.1	(19)
		MCWB	57.1	57.1	61.2	65.8	72.9	75.4	77.3	76.5	74.6	70.6	62.7	61.2	(20)
	2%	DB	59.8	62.6	70.8	82.3	88.0	93.5	96.4	93.5	89.3	81.1	69.4	62.0	(21)
		MCWB	53.6	51.7	57.4	64.4	71.5	73.9	76.1	75.4	72.6	68.9	59.3	56.4	(22)
	5%	DB	54.5	56.1	65.5	76.8	84.2	90.4	93.5	90.7	85.6	76.9	65.4	57.3	(23)
		MCWB	48.0	47.2	54.2	61.5	68.9	72.6	74.7	74.2	71.1	67.0	56.8	51.8	(24)
	10%	DB	49.8	51.2	60.9	71.8	80.3	87.3	90.9	88.2	82.6	73.2	62.0	53.1	(25)
		MCWB	43.4	43.2	51.1	58.9	66.9	71.4	74.0	73.0	69.8	64.3	54.7	47.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.4	60.1	63.4	69.2	75.4	77.7	80.1	79.8	77.8	74.8	65.4	62.7	(27)
		MCDB	64.8	66.3	74.4	81.2	89.3	91.3	95.0	92.0	87.2	79.7	71.2	65.5	(28)
	2%	WB	54.7	53.8	59.9	66.3	73.3	76.2	78.4	78.0	76.0	71.7	62.0	57.4	(29)
		MCDB	59.1	59.6	68.5	77.8	85.1	88.4	91.6	88.9	84.1	78.8	67.0	61.2	(30)
	5%	WB	48.9	48.2	56.2	63.8	71.2	75.1	77.2	76.6	74.5	69.1	59.1	52.5	(31)
		MCDB	53.6	55.1	63.6	74.5	81.6	86.4	89.0	86.3	81.9	75.8	63.8	56.2	(32)
	10%	WB	44.3	44.1	52.4	61.1	69.0	73.8	76.1	75.3	72.7	65.5	56.1	48.1	(33)
		MCDB	48.9	50.1	59.3	70.5	78.0	83.9	86.8	84.0	79.8	71.8	60.9	52.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	13.3	14.9	16.7	15.5	15.4	15.0	14.0	13.8	14.1	13.4	11.5	(35)
		MCDBR	19.1	20.6	21.8	23.6	19.7	18.6	17.7	17.1	17.3	17.9	17.6	16.5	(36)
		MCWBR	15.0	14.6	13.7	11.2	9.6	7.3	6.3	6.4	7.4	9.1	11.6	14.2	(37)
	5% WB	MCDBR	18.0	19.7	20.8	20.8	18.2	16.4	15.8	15.0	14.1	14.3	15.1	15.3	(38)
		MCWBR	15.6	15.9	15.0	11.3	9.2	7.3	6.4	6.5	7.5	9.2	12.2	14.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.312	0.322	0.362	0.401	0.446	0.501	0.519	0.514	0.430	0.365	0.326	0.311	(40)
	taud		2.465	2.419	2.374	2.285	2.193	2.083	2.049	2.046	2.253	2.441	2.506	2.505	(41)
	Ebn at Noon		273	285	283	278	266	251	245	242	257	266	266	265	(42)
	Edh at Noon		25	30	35	41	46	51	53	51	39	30	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		641	913	1210	1555	1738	1929	1910	1690	1379	1019	734	562	(44)
	RadStd		53	106	89	126	208	148	94	95	162	111	62	55	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	+131

Nomenclature: See separate page